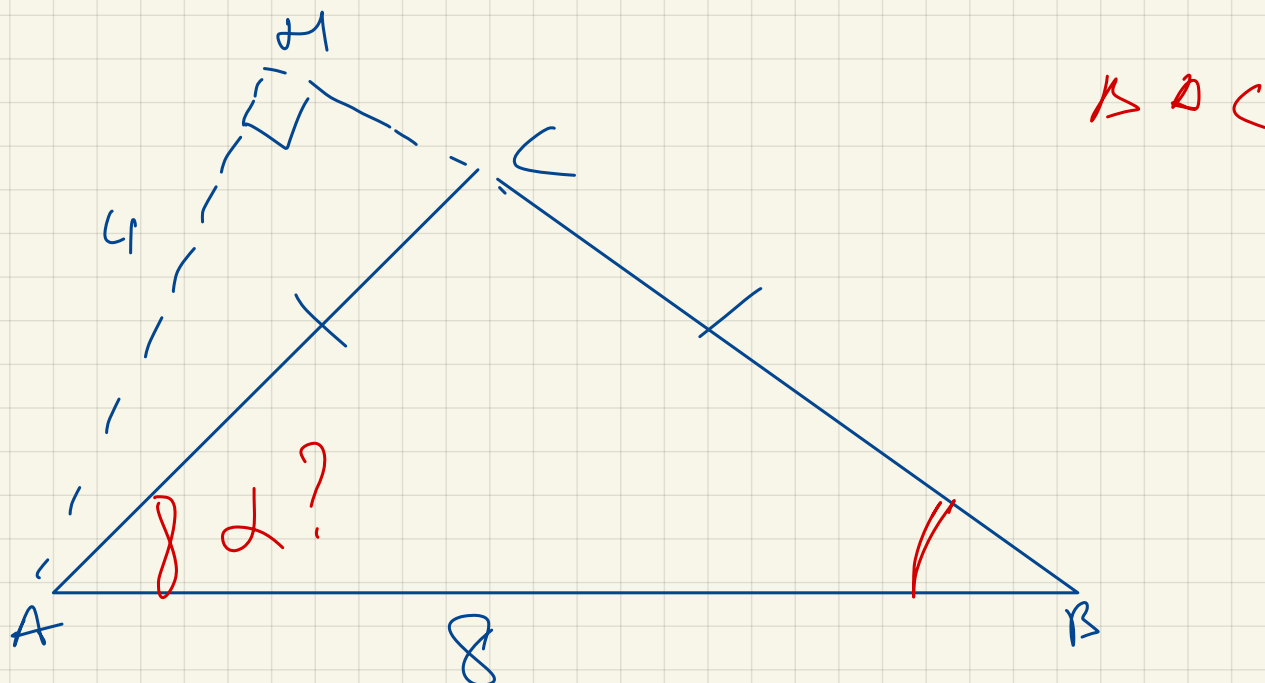
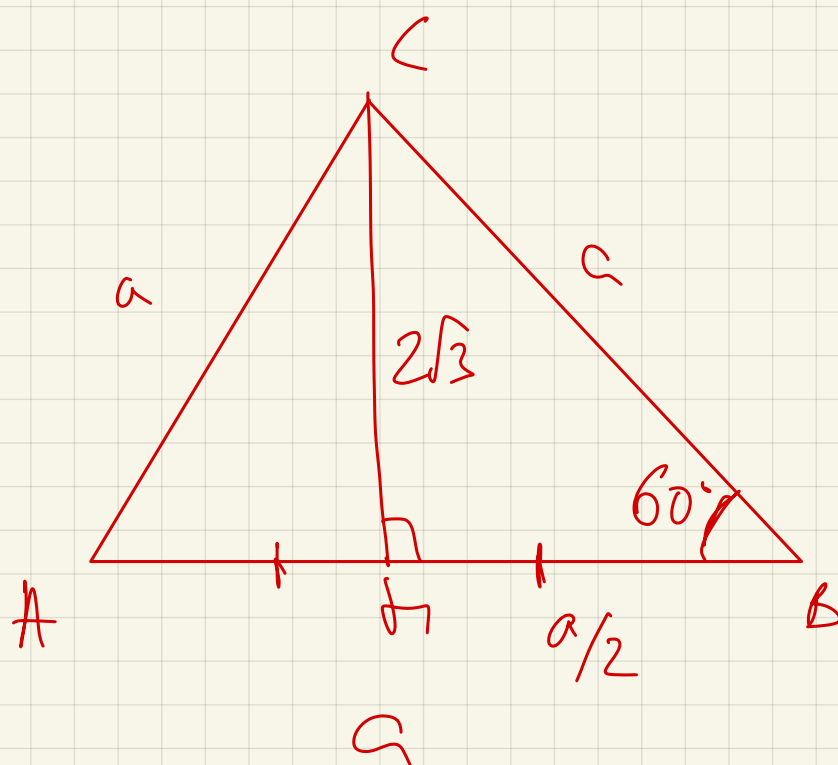






$$\sin B = \frac{AC}{AB} = \frac{4}{8} = \frac{1}{2}$$



a?

$$\frac{CH}{CB} = \sin 60^\circ$$

$$CB = \frac{CH}{\sin 60^\circ} =$$

$$\frac{2\sqrt{3}}{\sqrt{3}} = 4$$

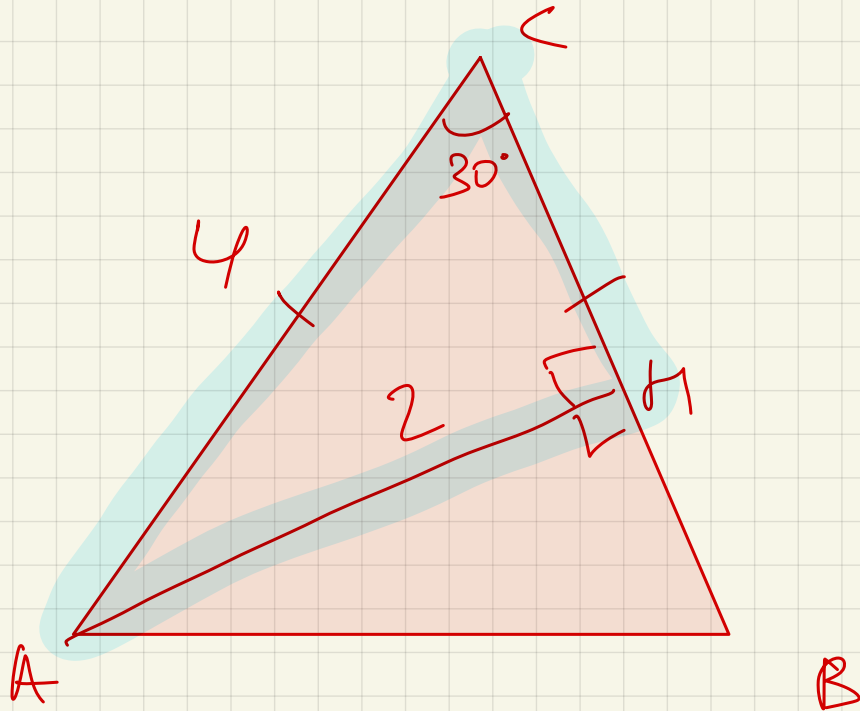
$$CB^2 = HB^2 + CH^2$$

$$a^2 = \left(\frac{a}{2}\right)^2 + CH^2$$

$$CH^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$$

$$CH = \frac{a\sqrt{3}}{2}$$

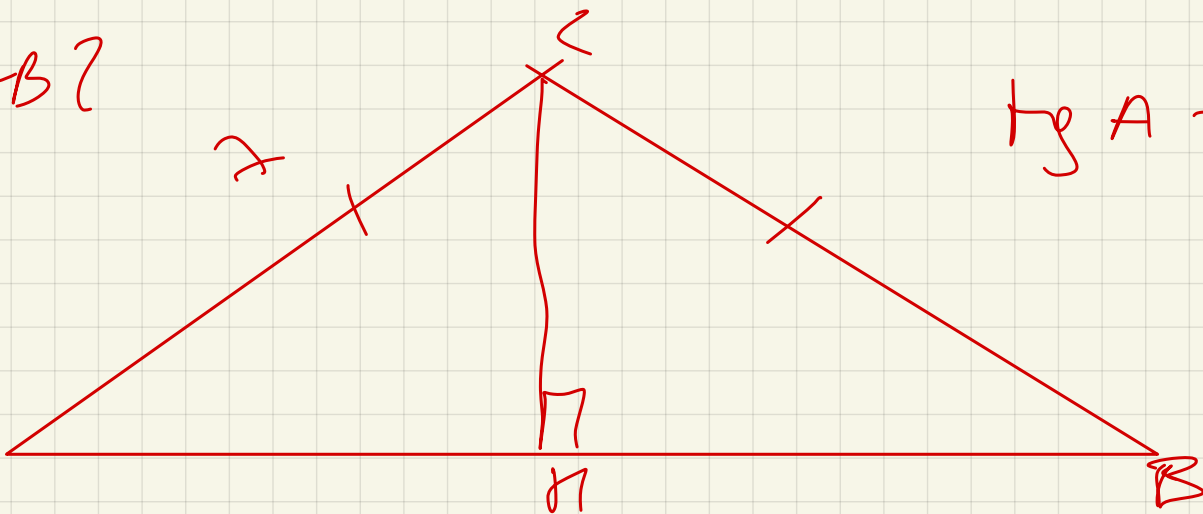
$$a = \frac{2 \cdot CH}{\sqrt{3}} = \frac{2 \cdot 2\sqrt{3}}{\sqrt{3}} = 4$$



$$\frac{AH}{AC} = \sin 30^\circ$$

$$AH = AC \cdot \sin 30^\circ = 4 \cdot \frac{1}{2} = 2$$

AB?



$$\tan A = \frac{3}{4\sqrt{3}}$$

$$\tan A = \frac{CH}{AH}$$

$$\tan A = CH$$

$$\left\{ \begin{array}{l} AC^2 + AH^2 = AC^2 \\ AH \lg A = CH \\ HC = AH \cdot \frac{33}{4\sqrt{33}} \end{array} \right.$$

$$AH^2 \cdot \left(\frac{33}{4\sqrt{33}} \right)^2 + \underline{AH^2} = AC^2 = 49$$

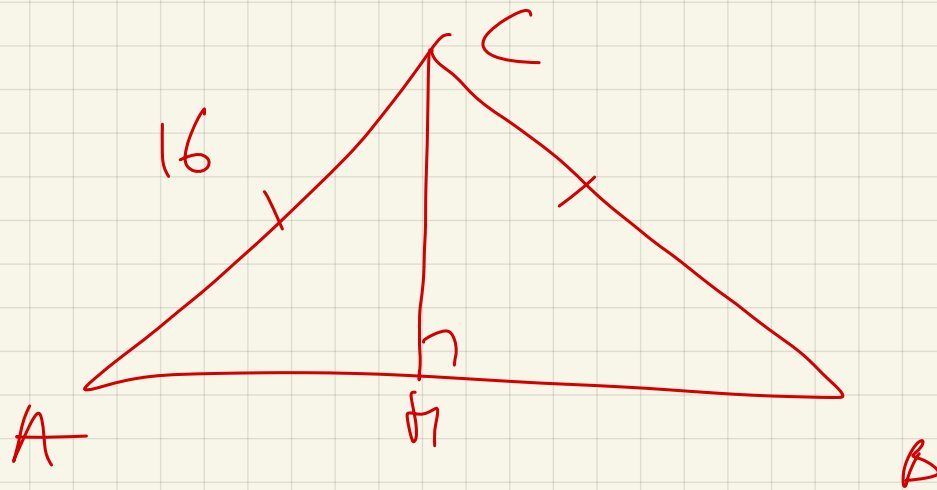
$$AH^2 \left(\frac{33^2}{4^2 \cdot \cancel{33}} + 1 \right)$$

$$AH^2 \left(\frac{33+16}{16} \right) = 49$$

$$AH^2 = \frac{16 \cdot \cancel{49}}{\cancel{49}}$$

$$\underline{AH = 4}$$

$$\underline{AB = 8}$$



$$\lg A = \frac{7}{3\sqrt{7}}$$

AB?

$$\frac{8}{g}$$

$$HC = \left(\frac{7}{3\sqrt{7}} \right)^2 \cdot \frac{4g^4}{g \cdot 7} = \frac{7}{9} + 3 =$$

$$\frac{16}{9} An^2$$

$$\frac{HC}{An} = \frac{7}{3\sqrt{7}}$$

$$HC = An \cdot \frac{7}{3\sqrt{7}}$$

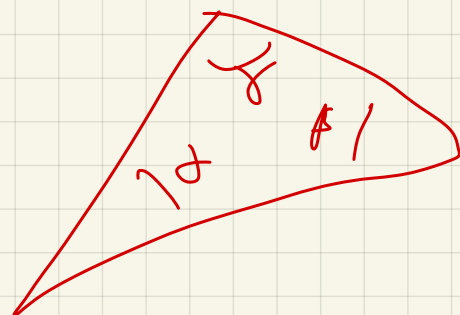
$$\frac{16}{9} An^2 = AC^2 \Rightarrow$$

$$\frac{16}{9} An^2 = 16^2$$

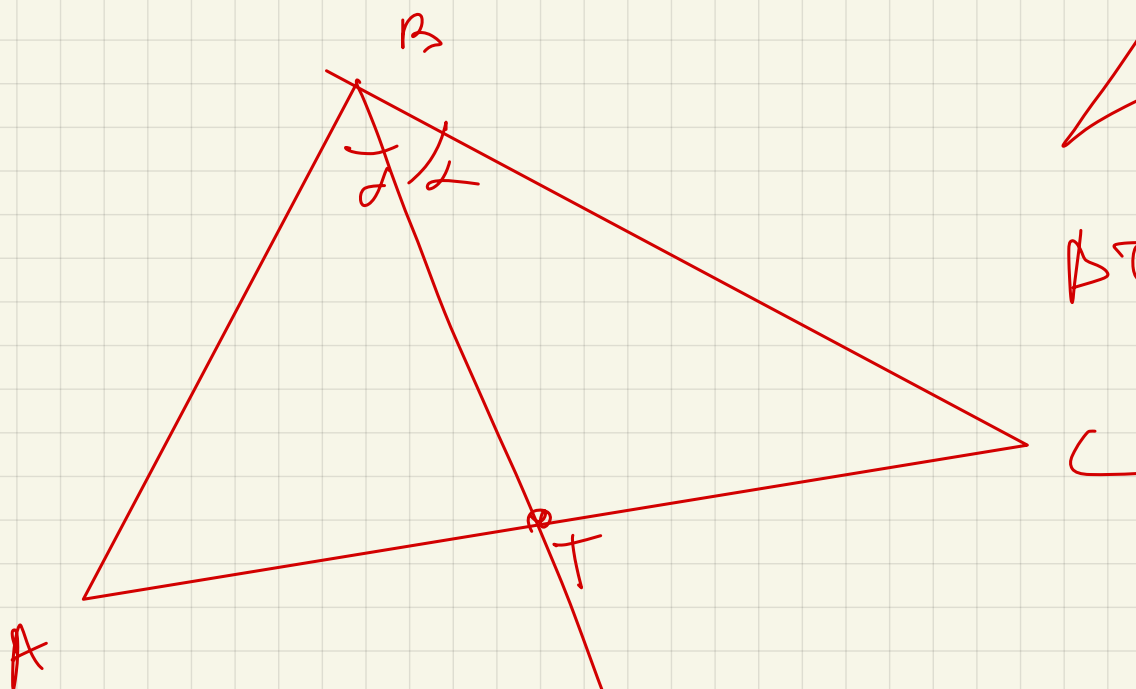
$$An^2 = \frac{16 \cdot 9}{16}$$

$$An^2 = 16 \cdot 9$$

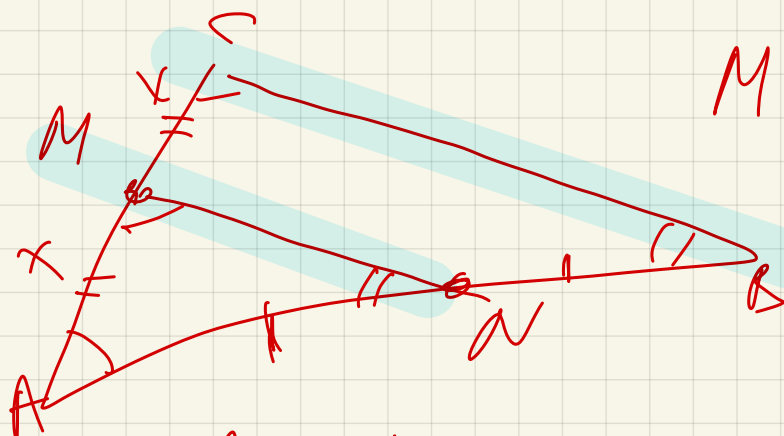
$$\sum = 180^\circ = \alpha + \beta + \gamma$$



BT - Тука.



$$\frac{BA}{AT} = \frac{BC}{TC}$$



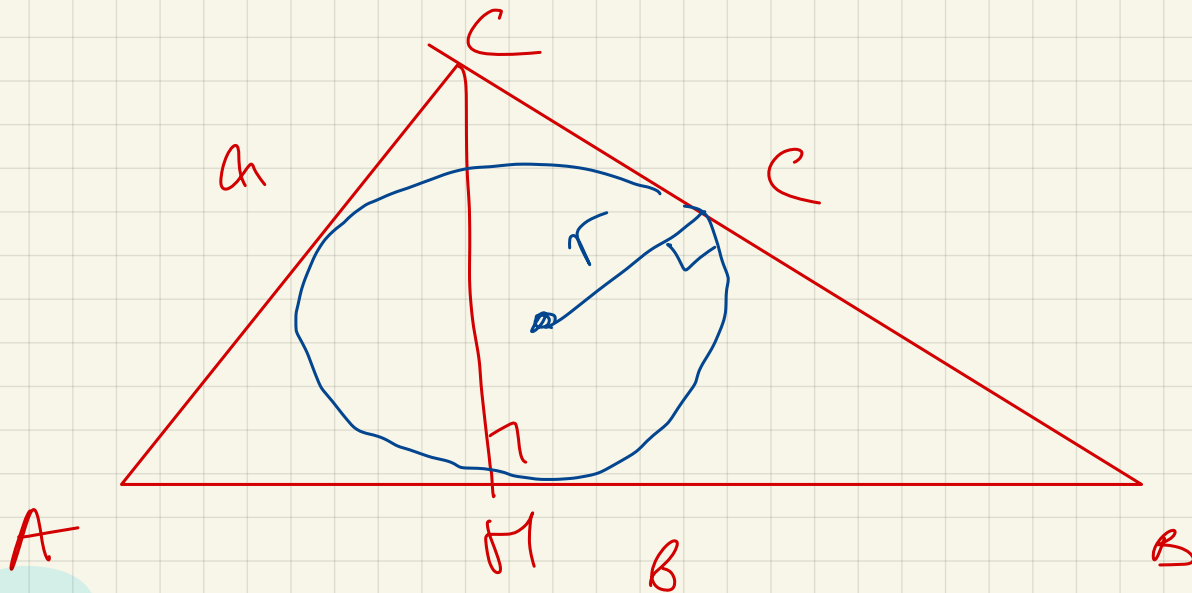
$$MN = \frac{CB}{2}$$

$$\triangle AMN \sim \triangle ACB$$

A - оодурт $\left. \begin{array}{l} \angle M = \angle C \\ \angle N = \angle B \end{array} \right\}$ нрм || нрмрм

$$S_{Hmn} = k^2 \cdot S_{ABC}$$

k - коэффициент.



$$S = \frac{CH \cdot AB}{2}$$

$$S = \frac{AC \cdot AB \sin A}{2}$$

$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

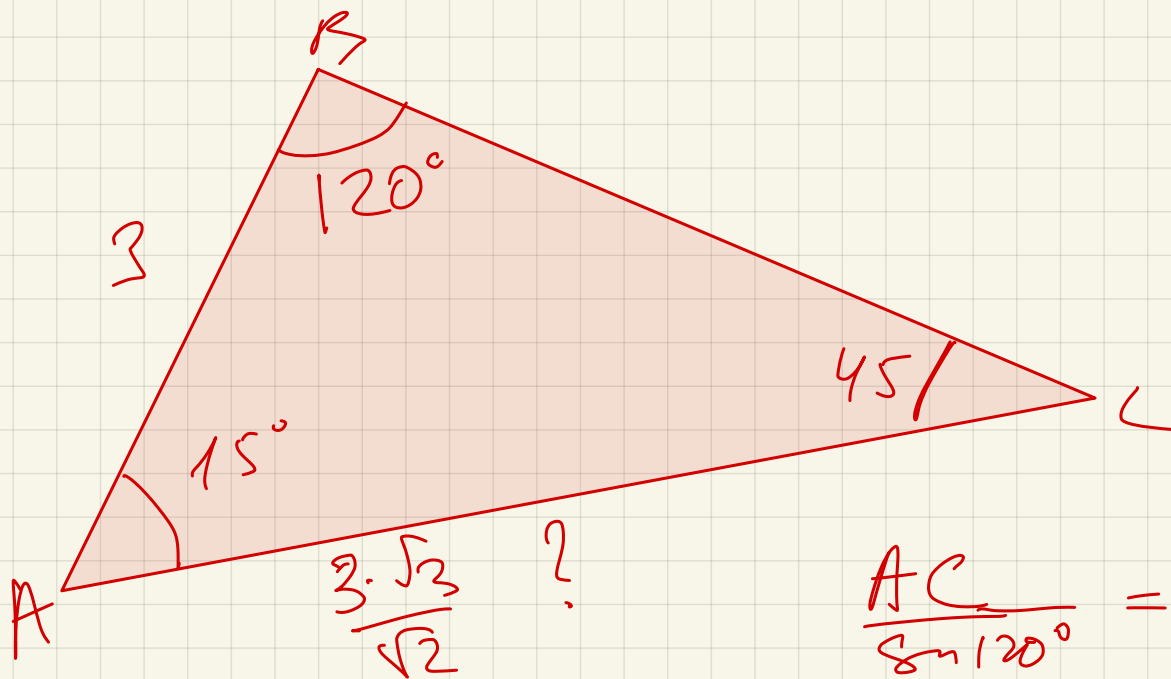
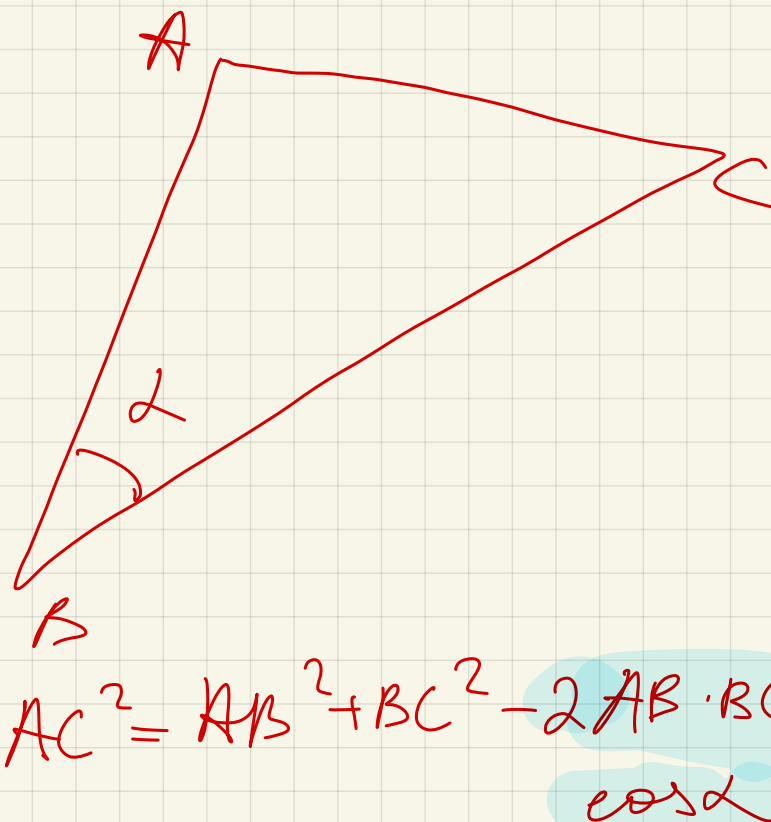
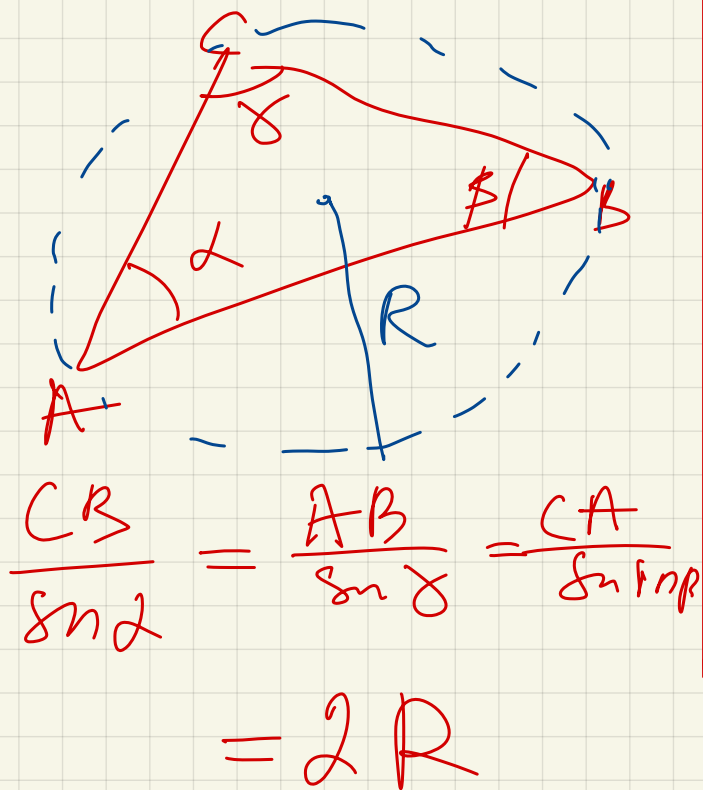
где

$$S = r \cdot p$$

$$p = \frac{a+b+c}{2}$$

Теорема Sin

Теорема Cos



$$\frac{AC}{\sin 120^\circ} = \frac{AB}{\sin 45^\circ}$$

BC?

S?

$$AC = \frac{AB}{\sin 45^\circ} \cdot \sin 120^\circ$$

∠BAC?

$$\frac{2AB}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} = AB \cdot \frac{\sqrt{3}}{\sqrt{2}}$$

$$\underline{\cos(45-30)} = \cos 45 \cos 30 + \sin 45 \sin 30$$

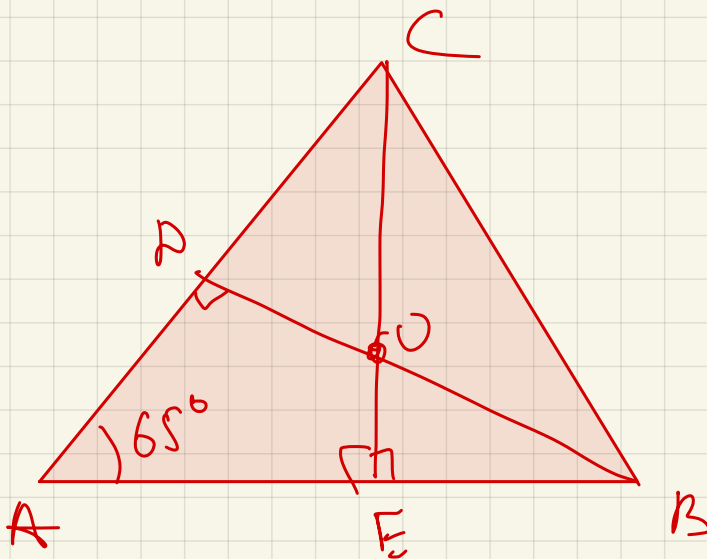
$$\frac{\sqrt{2}}{2} \quad \frac{\sqrt{3}}{2} \quad \frac{\sqrt{2}}{2} \quad \frac{1}{2}$$

$$\cos 15 = \left(\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} \right)$$

BC?

$$BC^2 = 3^2 + \left(\frac{3\sqrt{3}}{\sqrt{2}} \right)^2 - 2 \cdot 3 \cdot \frac{3\sqrt{3}}{\sqrt{2}} \left(\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} \right)$$

$$9 + \frac{9 \cdot 3}{2} - 2 \cdot 9 \frac{\sqrt{3}}{\sqrt{2}} \left(\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} \right)$$



$\angle AOE$?

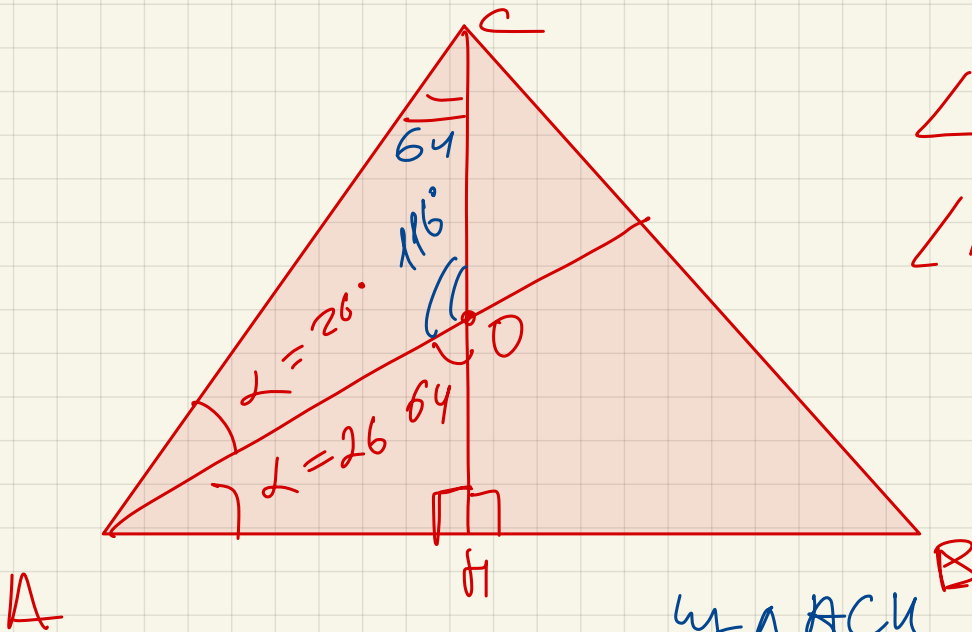
$$\angle D = 90^\circ$$

$$\sum_4 = 360^\circ$$

$$\begin{aligned} \angle O &= 360 - 90 - 90 - 65 \\ &= 180 - 65 = 115^\circ \end{aligned}$$

$\angle AOC?$

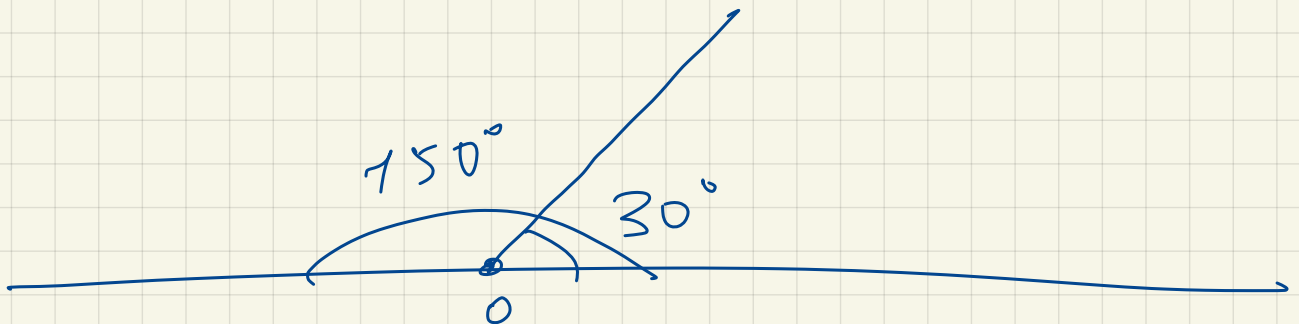
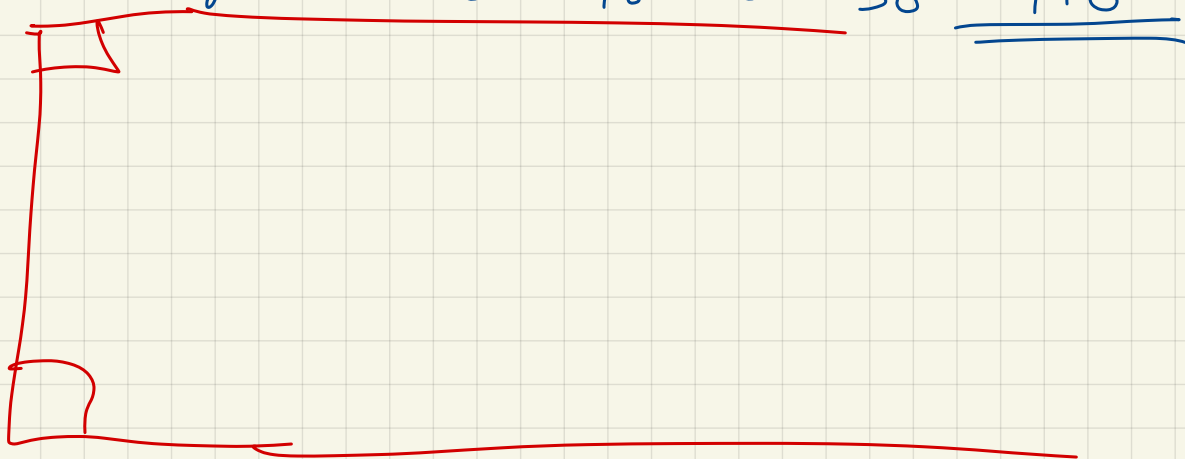
$\angle AOH = 64^\circ$



in $\triangle ACH$

$$\angle C = 180 - 90 - 26 - 26 = 38$$

$$\text{in } \triangle ACO = 180 - 64 - 38 = \underline{\underline{116}}$$

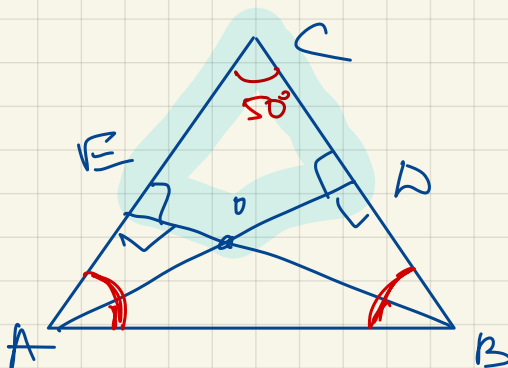


$\angle A$

58° u 72°

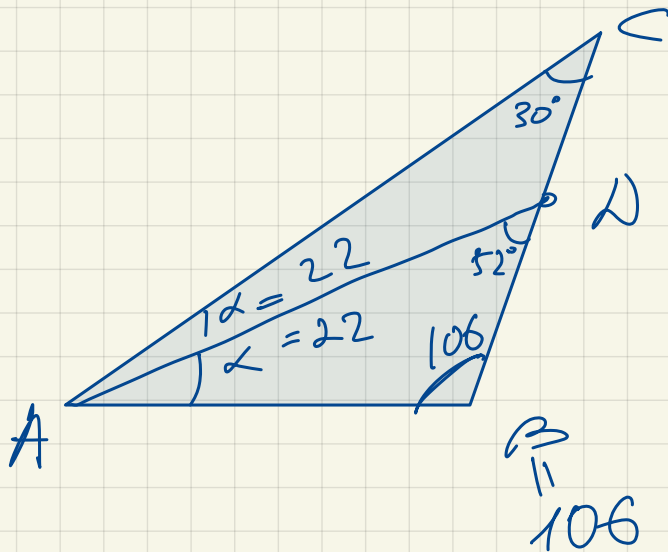
$\angle ADB?$

$\angle C = 50^\circ$

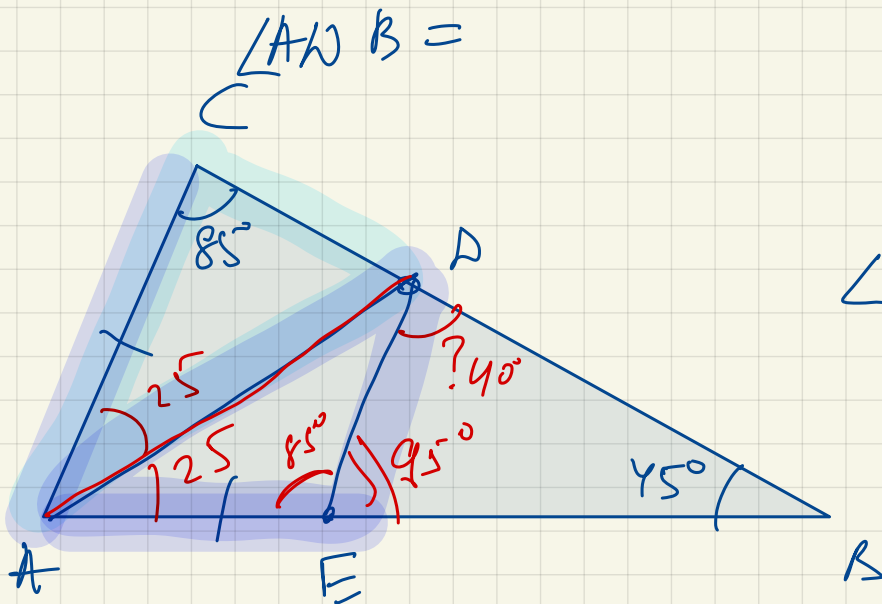


$AD = AC$?

52° ?



$$\angle B = 180 - 30 - 22 - 22 = 106$$



$AE = AC$

$\angle BDE$?

$\angle A = 50^\circ$

$\triangle CAD \sim \triangle DAE$
 $\angle DEB =$

